

WASP-104b

R-2659

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_150214 · created 2026-08-30T00:04:49+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457067.90058 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.014
Transit depth (Rp/Rs)^2	2.2 +/- 0.43 [%]
Orbital Inclination (inc)	82.68 +/- 0.91
Airmass coefficient 1 (a1)	1.002 +/- 0.011
Airmass coefficient 2 (a2)	-0.001 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	#1 - [506, 117]
Optimal Aperture	3.34
Optimal Annulus	4.61
Transit Duration (day)	0.067 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.014 vs 0.1214 (deviation 1.9σ)
Duration fit vs published	0.067 d vs 0.0742 d (deviation 0.72σ)
Mid-time O–C	1.1 min
Depth SNR	10.6
Residual scatter	1.14 %

Light curve

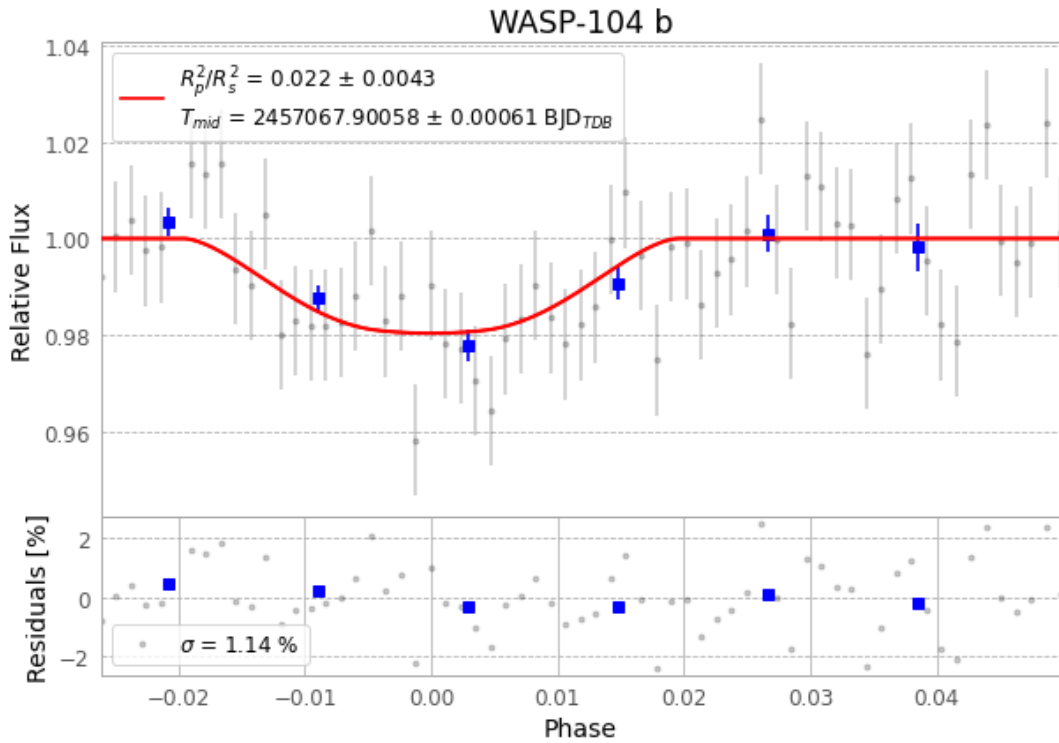


Figure 1 — FinalLightCurve_WASP-104 b_2015-02-14.png (SHA-256 4aacfdca608aafd5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e9083195bb2562a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), WASP-104150214082112.FITS → WASP-104150214113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-150214.log (349e8e90ce3c...), FinalLightCurve_WASP-104 b_2015-02-14.png (4aacfdca608a...), FinalLightCurve_WASP-104 b_2015-02-14.csv (8fbba593c0c8...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:04Z.